

Airbnb Global Performance Analysis

1. Business Problem

Airbnb needs to understand its global marketplace performance across cities, property types, hosts, pricing, customer reviews, and seasonal demand.

The business problem is to identify:

- Which cities contribute most to listings and reviews?
- How does property type affect pricing and marketplace performance?
- Which cities provide better customer experiences?
- What customer behavior can be understood from review frequency?
- How do seasonality and host verification influence marketplace activity and trust?

2. Business Questions

A. Marketplace Performance

1. What is the total number of Airbnb listings?
2. How many cities, hosts, and property types are represented?
3. Which cities contribute the highest number of listings?
4. What percentage of total listings and reviews comes from the top-performing cities?
5. How has the number of new listings changed over time?

B. Pricing and Property Types

1. What is the average price for each property type?
2. How do prices differ between entire places, private rooms, shared rooms, and hotel rooms?
3. Which property types contribute most to the marketplace?
4. Is there a noticeable pricing difference between cities and property types?

C. Customer Experience

1. Which cities have the highest overall ratings?
2. Which cities have comparatively lower ratings?
3. Which rating categories require improvement?
4. How do cities compare in accuracy, cleanliness, communication, location, and value for money?
5. Is value for money a potential concern across the marketplace?

D. Customer Review Behavior

1. How frequently do customers submit reviews?
2. What percentage of reviewers submit only one or a few reviews?
3. Are there customers with unusually high review activity?
4. How does review activity change across different months?
5. Which cities experience higher review activity during peak travel periods?

E. Host Trust and Verification

1. What percentage of hosts have verified identities?
2. How many hosts provide profile pictures?

3. What proportion of hosts have no profile picture or unverified identity?
4. What trust signals are available to customers before booking?
5. How strong is the overall host verification level?

3. KPIs Used and Their Purpose

KPI / Measure	Why It Was Used
Total Listings	Measures the overall size of Airbnb's marketplace.
Total Cities	Shows the geographic coverage of the analysis.
Total Hosts	Measures the number of supply-side participants.
Total Property Types	Shows the variety of accommodation options.
Total Reviews	Indicates the volume of customer feedback available.
New Listings	Measures marketplace growth over time.
Listings by City	Identifies major markets and geographic concentration.
Review Share by City	Shows which cities generate the most customer engagement.
Average Price by Property Type	Compares pricing across accommodation categories.
Overall Rating	Measures the general customer experience.
Accuracy Rating	Evaluates how accurately listings represent the actual property.
Cleanliness Rating	Identifies potential service-quality issues.
Communication Rating	Measures host responsiveness and communication quality.
Location Rating	Evaluates customer satisfaction with property location.
Value-for-Money Rating	Measures whether customers consider the price reasonable.
Review Frequency	Helps understand customer engagement and repeat reviewing behavior.
Monthly Review Share	Identifies seasonal demand and travel patterns.
Host Verification Rate	Measures the strength of identity-based trust signals.
Profile Picture Rate	Indicates the visibility and completeness of host profiles.

4. Why These Measures Were Selected

1. **Marketplace KPIs** were used to understand Airbnb's size, reach, and supply distribution.
2. **City-level measures** were used to identify important markets and concentration of listings and reviews.
3. **Pricing measures** were used to compare accommodation categories and understand price differences.
4. **Rating measures** were used to evaluate customer experience and identify service-quality gaps.
5. **Review measures** were used to understand customer engagement and seasonal activity.
6. **Verification measures** were used to evaluate host transparency and customer trust.

5. Main Dashboard Insights

- The dataset contains **279,712 listings**, **182,024 hosts**, **10 cities**, **144 property types**, and approximately **5.373 million reviews**.
- Paris, New York, and Sydney contribute almost half of the total listings and approximately **59% of total reviews**.
- Paris has the highest number of listings and reviews among the analyzed cities.
- Mexico City and Rio de Janeiro have the highest overall ratings.
- Cleanliness and value for money are the comparatively weaker rating dimensions.
- Approximately **98.8% of reviewers** submitted reviews three times or fewer.
- Paris and Rome show higher review activity between April and August.
- More than two-thirds of hosts are fully verified.
- Airbnb recorded its highest number of new listings in 2015, followed by slower growth and disruption during the COVID-19 period.